

Prediction of Diabetes in Middle-Aged Adults: A Machine Learning Approach

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Abstract

Background: Diabetes is a serious health concern requiring effective diagnostic strategies, particularly since its symptoms overlap with those of other conditions. Despite extensive research on early diabetes detection across various age groups, middle-aged adults have been relatively underexplored. This study focuses on this demographic to examine symptom-diabetes associations, examine the influence of symptoms in diabetes prediction, and determine an optimal machine learning (ML) model for diabetes prediction. **Materials and Methods:** This study utilized data from a previous cohort study conducted in Bangladesh. The original dataset included demographic and symptom-related information from 520 patients visiting the ABC Hospital in Bangladesh, India. The participants comprised both diabetic and non-diabetic individuals showing diabetes-like symptoms. For our study, data from 296 middle-aged adults (aged 40–60 years) were extracted. Chi-square tests assessed diabetes-symptom associations, and the Boruta algorithm examined feature influence. Seven ML classification models were evaluated for predictive accuracy. **Results:** Results showed that 60% of the 296 participants were diabetic. Symptoms like polyuria, polydipsia, weakness, sudden weight loss, partial paresis, polyphagia, and visual blurring were significantly associated with diabetes. All demographic and symptom-related features were influential in diabetes prediction, with polyuria, polydipsia, gender, alopecia, and irritability emerging as the most influential. Among the ML models tested, the random forest model exhibited the highest sensitivity (98.59%) and outperformed others in accuracy (96.58%) and area under the curve score (96.00%), making it the most efficient model for predicting diabetes in middle-aged adults. **Conclusion:** Diabetes associated symptoms provide valuable diagnostic opportunities for early diabetes detection in middle-aged adults. Future research should explore genetic, lifestyle, and environmental factors to improve diagnostic accuracy.

Keywords: Diabetes prediction, diabetes symptoms, machine learning models, middle-aged adults, predictive performance metrics

Key Messages: Our study reveals that diabetes-associated symptoms provide valuable diagnostic opportunities, enabling accurate diabetes predictions in middle-aged adults.

INTRODUCTION

Diabetes stands as a persistent medical condition marked by heightened blood glucose levels and disruptions in the metabolism of fats and proteins.^[1] This chronic ailment encompasses a cluster of symptoms linked to hyperglycemia, signifying elevated sugar levels in the bloodstream.^[2] The complex process by which the

body converts food into energy is altered in diabetes. Typically, the body breaks down food into sugar and glucose, releasing them into the bloodstream. Elevated blood sugar levels prompt the pancreas to release insulin, acting as a key to facilitate the entry of sugar into the

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body's cells for energy utilization.^[3] In diabetes, either insufficient insulin is produced, or the body fails to use insulin effectively, leading to excessive blood sugar lingering in the bloodstream. Over time, this condition can give rise to severe health complications such as heart disease, vision impairment, and kidney disorders.^[3] The global prevalence of diabetes has surged, quadrupling since 1980, with an estimated 422 million people worldwide affected, according to the World Health Organization.^[4]

The Centers for Disease Control and Prevention identifies three primary types of diabetes: type 1 (T1D), type 2 (T2D), and gestational (GD) diabetes. T1D is a heterogeneous disorder characterized by the destruction of pancreatic beta cells, resulting in absolute insulin deficiency.^[5] As an autoimmune disorder, T1D is considered the most common and severe among autoimmune diseases,^[6] though it accounts for only 5%–10% of global diabetes cases.^[7,8] T1D can manifest at any age, with the highest incidence observed in children.^[9] On the other hand, T2D is a progressive disorder marked by deficiencies in insulin secretion and increased insulin resistance, leading to abnormal glucose metabolism and related metabolic disruptions.^[10] Representing approximately 90% of all diabetes cases, T2D predominantly affects middle-aged adults.^[11] Symptoms of T2D may develop gradually over several years, sometimes going unnoticed for an extended period.^[12] Lastly, GD is a less common form of diabetes occurring in pregnant women without preexisting diabetes, leading to elevated blood sugar levels during pregnancy. The outcomes of GD include an elevated risk of maternal cardiovascular disease and T2D.^[13] Additionally, infants born to mothers with GD may experience macrosomia (excessive birth weight) and complications during birth.^[13] Globally, 14% of all pregnancies are affected by GD.^[14]

Irrespective of the diabetes type, neglecting early signs and symptoms can result in severe long-term complications, including cardiovascular issues, kidney problems, vision impairment, and neuropathic conditions. Recognizable early signs that necessitate diabetes diagnosis and investigation include frequent urination (polyuria), excessive thirst (polydipsia), sudden weight loss, excessive hunger (polyphagia), blurred vision, unexpected weakness, dry skin, delayed healing, and frequent infections.^[15] Given the overlap of symptoms with other medical conditions, there is a crucial need for highly efficient and user-friendly diagnostic methods to aid healthcare professionals and patients in determining the diabetes status of individuals presenting with similar symptoms.

In light of the escalating global incidence and mortality rates of diabetes,^[16] numerous studies have delved into understanding the etiology, diagnosis, and management of the condition. For the effective management of diabetes,

early detection is crucial, as the condition can deteriorate over time if not properly addressed or remains undetected. While many diagnostic studies rely on laboratory-based tests, some are outdated^[17–19] despite some having high accuracy rates.^[19] The recent surge in artificial intelligence has prompted a growing number of studies utilizing machine learning (ML) techniques to detect and screen diabetes across various demographics and populations, including gender,^[20–22] income classes,^[23] and age groups such as infants,^[24] children, and teenagers.^[25–27] Surprisingly, few, if any, of these studies explicitly concentrate on employing ML techniques to screen and diagnose diabetes in middle-aged adults. This study aims to fill this gap by focusing on using ML methods to detect diabetes, specifically in the middle-aged adult population.

The primary objectives of this study are to (1) assess associations between symptoms displayed by middle-aged adults and their diabetes status, (2) investigate the relevance and relative influence of some demographic and symptomatic features in predicting diabetes status among middle-aged adults, and (3) identify the most effective ML model for predicting diabetes status in middle-aged adults. The significance of this study lies in its potential to provide valuable insights into the associations between symptoms, demographic factors, and diabetes status in the middle-aged population.

MATERIALS AND METHODS

Study design and data source

This research utilized the dataset derived from a previous study conducted by Islam *et al.*^[28] in India, where they aimed at constructing a predictive tool for early-stage diabetes risk predictions. The dataset, publicly available on Mendeley Data,^[29] includes information on patient demographics and diabetes-related symptoms obtained from 520 participants through a questionnaire. These participants were patients visiting the Sylhet Diabetes Hospital in Bangladesh and included both diabetic and non-diabetic individuals showing signs similar to that of diabetes. The dataset contains comprehensive information on patient symptoms, demographics, and diabetes status, making it suitable for addressing the objectives of this current investigation.

For our study, we specifically extracted data for 296 middle-aged adults within the original dataset, where the inclusion criterion was for individuals aged between 40 and 60 years. The exclusion criterion was for all other patients who fell outside this age bracket.

Features and feature selection

Within the dataset employed for this study, 17 variables were identified, with 16 serving as features and one as the outcome variable. The outcome variable contained information about a patient's diabetes status, while the

features included details on patient demographics and symptoms linked to diabetes, as outlined in Table 1. All features, excluding the “age” variable, were nominal with two classes. Notably, our dataset contained no missing values.

To assess the influence of features on predicting diabetes status, we employed the Boruta algorithm, accessible through the Boruta package in R,^[30] for feature selection. The Boruta algorithm, recognized for its all-relevant approach, utilizes a wrapper algorithm to identify influential features. Leveraging the random forest (RF)

algorithm, it conducts a comprehensive search for relevant features by comparing original variable importance with estimates derived from permuted copies, progressively eliminating irrelevant features. This algorithm is adaptable, accommodating any classification method outputting variable importance.

For a visual assessment of feature influence and relevance, we utilized a Boruta algorithm-generated plot, illustrated in Figure 1. In this plot, the vertical axis represents feature importance, while the horizontal axis represents features. The color of each boxplot indicates the relevance of the feature in predicting the outcome variable. Blue boxplots correspond to the minimal, average, and maximum Z score of a shadow feature, while red, green, and yellow boxplots represent Z scores of rejected, confirmed, and tentative features, respectively. Confirmed features are those recognized as significant predictors of the outcome variable. Rejected features, on the other hand, are deemed non-important in predicting the outcome variable. Tentative features are predictors for which the Boruta algorithm could not conclusively determine their importance.

Table 1: Overview of the features used in this research to study diabetes among middle-aged adults

Feature	Description
Age	Age of patient
Sex	Gender of patient
Polyuria	Signs of frequent urination
Polydipsia	Signs of extreme thirstiness
Polyphagia	Signs of extreme hunger
Sudden weight loss	Unexplained weight loss
Genital thrush	Recent genital yeast infection
Visual blurring	Blurred vision or poor eyesight
Itching	Frequent itching of the body
Irritability	Feeling agitated or frustrated
Delayed healing	Poor healing of wounds
Partial paresis	Partial loss of voluntary movement in the limbs
Muscle stiffness	Stiff muscles
Alopecia	Sudden unexplained hair loss
Obesity	Obese patient
Diabetes status	Diabetes status of the patient

Statistical analysis

To prepare our data for analysis, we conducted data preprocessing. Except for the “age” variable, all other variables were nominal with two classes. Utilizing one-hot encoding, we encoded all nominal variables, representing the presence of a category with “1” and its absence with “0.” Additionally, we applied the min-max scaling technique to standardize the “age” variable’s scale, bringing its values within the range of 0 to 1.

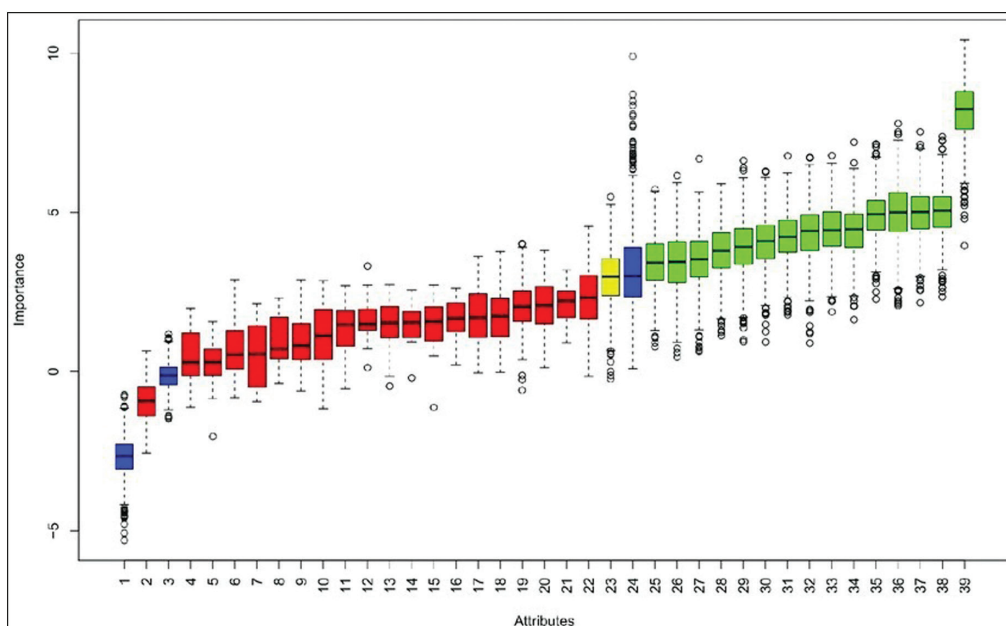


Figure 1: Illustration of a sample Boruta algorithm-generated feature importance plot

In this study, descriptive statistics were employed to characterize the primary outcome variable and all other predictor variables. Frequencies and percentages were used to describe the nominal variables. To examine associations between symptoms exhibited by middle-aged adults and their diabetes status, we employed the Chi-square test of association. This statistical method evaluates the dependence or independence between two nominal variables, determining whether a significant association exists based on observed and expected frequencies within a contingency table.

Furthermore, to determine the most efficient ML algorithm for predicting diabetes within the middle-aged adult population, we assessed the performance of seven distinct algorithms. These algorithms include K-nearest neighbor (KNN), naïve Bayes classifier, support vector machines (SVM) with linear, polynomial, and radial basis function kernels, RF classifier, and logistic regression (LR). We employed 10-fold cross-validation to select the best-performing model for each type of ML algorithm to ensure robust model evaluation and minimize the risk of overfitting. In evaluating the performance of the ML models, various performance metrics were utilized, including accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and receiver operating characteristic (ROC) curve analysis with their associated area under the curve (AUC) values. In this study, the selection of the optimal ML model for predicting diabetes status among middle-aged adults was based on the one demonstrating both the highest accuracy and AUC scores.

The chosen level of significance throughout the study was set at 0.05. All statistical analyses were carried out using the R programming language (version 4.3.2, R Core Team, Vienna, Austria).

Ethical Approval

We did not seek ethical approval from the Committee on Human Research and Publication Ethics (CHRPE) at the School of Medicine and Dentistry, Kwame Nkrumah University of Science and Technology, because the data utilized in this study was secondary and publicly available under a Creative Commons Attribution 4.0 International license. The dataset is anonymized, containing no personally identifiable information, and we were not involved in the original data collection process. In compliance with the principles outlined in the Declaration of Helsinki, we ensured that all ethical standards regarding research on human data were maintained throughout the study.

RESULTS

Assessing the associations between symptoms displayed by middle-aged adults and their diabetes status

Table 2 displays symptoms in middle-aged patients, both with and without diabetes, along with the *P* value from Chi-square tests assessing associations with diabetes status. Among the 296 participants, 179 (60%) were diabetic. Symptoms like polyuria, polydipsia, weakness, sudden weight loss, partial paresis, polyphagia, and visual blurring are more prevalent in diabetic middle-aged patients and rare in their non-diabetic counterparts. This emphasizes the significance of these symptoms as potential indicators of diabetes in this age group.

The *P* value from the Chi-square tests highlights strong associations between diabetes status and symptoms like polyuria, polydipsia, weakness, sudden weight loss, partial paresis, polyphagia, and visual blurring in middle-aged individuals. These results emphasize the importance of these symptoms as strong indicators of diabetes in this age group. In contrast, symptoms such as obesity, delayed

Table 2: Summary results of the Chi-square tests of association between diabetes status and symptoms among middle-aged adults

Symptom	Diabetes (<i>n</i> = 179)		Diabetes free (<i>n</i> = 117)		<i>P</i> value
	Yes (%)	No (%)	Yes (%)	No (%)	
Polyuria	136 (76)	43 (24)	4 (3)	113 (97)	0.000
Polydipsia	134 (75)	45 (25)	8 (7)	109 (93)	0.000
Sudden weight loss	125 (70)	54 (30)	23 (20)	94 (80)	0.000
Weakness	132 (74)	47 (26)	64 (55)	53 (45)	0.001
Polyphagia	107 (60)	72 (40)	29 (25)	88 (75)	0.000
Genital thrush	43 (24)	136 (76)	30 (26)	87 (74)	0.859
Visual blurring	103 (58)	76 (42)	36 (31)	81 (69)	0.000
Itching	84 (47)	95 (53)	74 (63)	43 (37)	0.008
Irritability	53 (30)	126 (70)	7 (6)	110 (94)	0.000
Delayed healing	85 (47)	94 (53)	62 (53)	55 (47)	0.420
Partial paresis	114 (64)	65 (36)	22 (29)	95 (71)	0.000
Muscle stiffness	75 (42)	104 (58)	40 (34)	77 (66)	0.000
Alopecia	34 (19)	145 (81)	77 (66)	40 (34)	0.000
Obesity	39 (22)	140 (78)	23 (20)	94 (80)	0.769

healing, and genital thrush showed no association with diabetes status in middle-aged individuals, suggesting that these conditions may not be as specific to diabetes in this age group.

Investigating the relevance and relative influence of the demographic and symptomatic features in the prediction of diabetes status

Figure 2 visually presents the relevance and relative importance of each variable employed in our study to predict diabetes status in middle-aged adults. The plot reveals clinical and statistical implications that can enhance the understanding and application of these symptoms in the diagnosis of diabetes in this specific population.

From the plot, we observe that certain predictors, namely polyuria, polydipsia, gender, alopecia, irritability, and sudden weight loss, are highly influential features in descending order of importance when it comes to predicting the diabetes

status of middle-aged patients. Conversely, features such as genital thrush, itching, obesity, and muscle stiffness are shown to be less important when predicting diabetes status in middle-aged patients. This finding suggests that while these features still play a role in diabetes diagnosis, their relative importance is lower. The green-colored boxplots associated with all features signify that the Boruta algorithm identified each variable as an important contributor to predicting diabetes status in middle-aged patients.

Identifying the most effective ML model for predicting diabetes in middle-aged adults

Table 3 presents a comprehensive evaluation of the performance of the seven ML models used in predicting diabetes in our study, considering their sensitivity, specificity, PPV, NPV, and accuracy scores. From the table, the RF model demonstrated the highest sensitivity at 98.59%, outperforming other models, while KNN

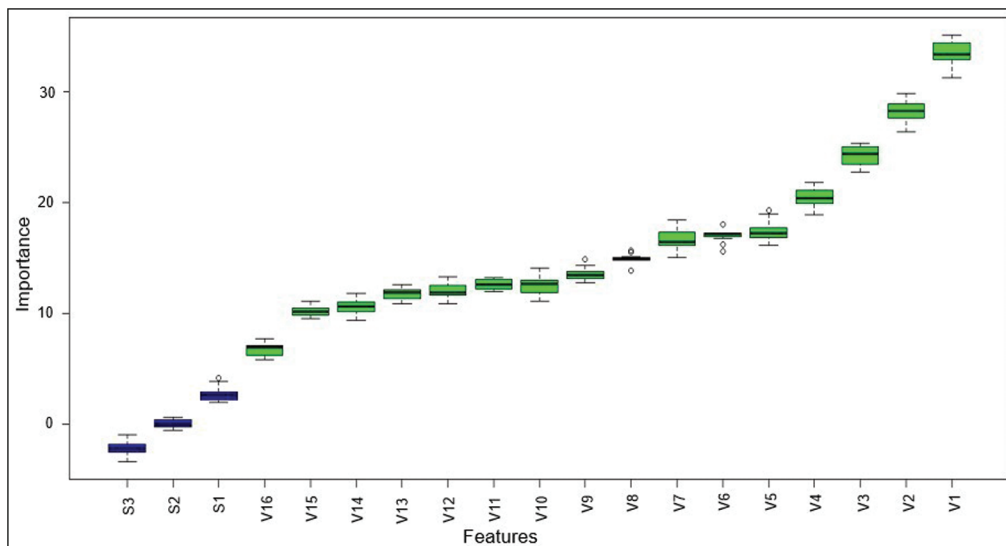


Figure 2: Relative importance of features in predicting diabetes status among middle-aged adults. V1 = polyuria, V2 = polydipsia, V3 = gender, V4 = alopecia, V5 = irritability, V6 = sudden weight loss, V7 = age, V8 = partial paresis, V9 = polyphagia, V10 = delayed healing, V11 = visual blurring, V12 = muscle stiffness, V13 = weakness, V14 = itching, V15 = obesity, V16 = genital thrush, S1 = ShadowMax, S2 = ShadowMean, S3 = ShadowMin

Table 3: Performance metrics associated with each machine learning model in the prediction of diabetes among middle-aged adults

Model	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	Accuracy (%)
KNN ($k = 5$)	85.92	97.83	98.39	81.82	90.60
NB	97.18	78.26	87.34	94.74	89.74
SVM (linear)	91.55	95.65	97.01	88.00	93.16
SVM (polynomial)	92.96	95.65	97.06	89.80	94.02
SVM (RBF)	91.55	93.48	95.59	87.76	92.31
RF	98.59	93.48	95.89	97.73	96.58
LR	92.96	95.65	97.06	89.80	94.02

KNN = K-nearest neighbor, NB = naïve Bayes, SVM (linear) = support vector machines with linear kernel, SVM (polynomial) = support vector machines with polynomial kernel, SVM (RBF) = support vector machines with radial basis function kernel, RF = random forest, LR = logistic regression, PPV = positive predictive value, NPV = negative predictive value

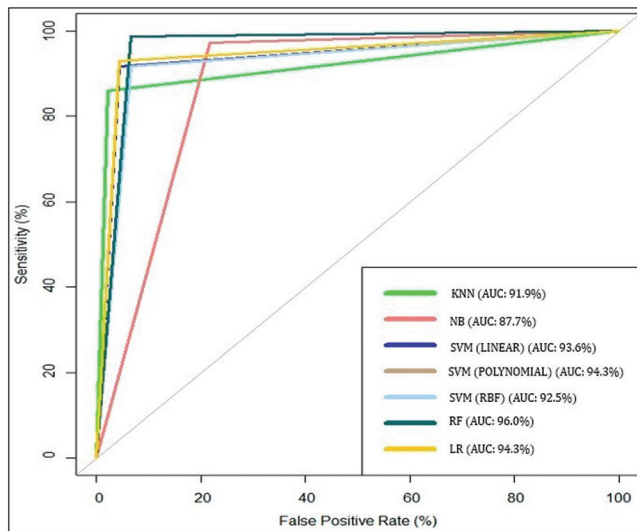


Figure 3: Receiver operating characteristic curves with their corresponding area under the curve scores of the machine learning models of this study. KNN = K-nearest neighbor, NB = naïve Bayes, SVM (linear) = support vector machines with linear kernel, SVM (polynomial) = support vector machines with polynomial kernel, SVM (RBF) = support vector machines with radial basis function kernel, RF = random forest, LR = logistic regression, AUC = area under the curve

ranked highest in specificity with a score of 97.83%. The KNN model achieved the highest PPV at 98.39%, indicating its reliability in identifying diabetic cases, while the RF model had the highest NPV at 97.73%, indicating its reliability in identifying non-diabetic cases. Overall accuracy was highest in the RF model with a score of 96.58%, emphasizing its effectiveness in predicting both diabetic and non-diabetic cases among middle-aged adults.

ROC and AUC analysis

Figure 3 illustrates the ROC curves and their corresponding AUC values for each model used in predicting diabetes among middle-aged adults in this study. AUC values are crucial indicators of a model's ability to distinguish between positive and negative cases. From the plot, the RF model stood out with an impressive AUC of 96.0%, surpassing other models. While AUC values were closely similar, SVM (polynomial) and LR had overlapping curves with identical AUC values of 94.3%. In summary, the RF model exhibited superior discriminative performance, making it the optimal choice for predicting diabetes in middle-aged adults. The consistently high AUC scores of the models highlight the robust predictive capabilities of ML models in the prediction of diabetes.

DISCUSSION

Diabetes is a significant health concern, and its symptoms can remain subtle or unnoticed for extended periods, particularly in T2D.^[16,31] Since key diabetes symptoms may

resemble those of other medical conditions, individuals often prioritize addressing those conditions. Consequently, diabetes diagnosis tends to occur in later disease stages when symptoms become more severe. This study explored the associations between diabetes status and specific symptoms in middle-aged adults, including both diabetic and non-diabetic individuals. Our findings indicate that among middle-aged adults, the prevalent symptoms associated with diabetes include polyuria, polydipsia, weakness, sudden weight loss, polyphagia, partial paresis, and visual blurring. Conversely, these symptoms are less common among their non-diabetic counterparts.

Several studies across different populations have reported findings that are consistent with ours.^[31-34] Devi *et al.*,^[31] in their investigation into the prevalence of T2D among the Meiteis of Manipur in India, identified similar symptoms among T2D patients, including polyuria, polyphagia, polydipsia, fatigue, and weight loss, all of which were significantly associated with diabetes. Similarly, Kumar and Kaplowitz^[32] observed a parallel discovery in a different age demographic, noting that children with diabetes commonly exhibited symptoms such as polyuria, polydipsia, weight loss, and fatigue. In another study, Pawar *et al.*^[33] found a strong indication of diabetes in patients displaying polydipsia, polyuria, polyphagia, or a combination of these symptoms. The consistency of our study's findings with those of previous studies in different populations suggests that middle-aged adults exhibit symptoms of diabetes similar to those in other demographics, with the typical symptoms being polyuria, polyphagia, polydipsia, visual blurring, and fatigue.

In contrast, our study revealed that symptoms such as obesity, delayed healing, and genital thrush were not associated with the diabetes status of middle-aged adults. This contradicts previous research that established a strong association between T2D and obesity.^[35-37] For instance, a 2011 study by Barnes^[37] found a significant increase in the risk of diabetes among overweight and obese women, with overweight women facing a 28-fold risk and obese women an astonishing 93-fold risk compared to normal-weight women. The reasons for these discrepancies between our study's findings and existing research are not entirely clear. One plausible explanation is that a majority of our study participants may be suffering from T1D, which is typically not associated with obesity.

Moreover, our study found no association between delayed wound healing and diabetes among middle-aged adults, contradicting previous research.^[38,39] Fahey *et al.*^[38] concluded in their study that delayed healing in diabetes is linked to altered leucocyte infiltration and wound fluid interleukin-6 levels during the late inflammatory phase of the wound healing process. Our findings suggest that certain symptoms traditionally associated with diabetes may not always be indicative in this specific demographic.

Therefore, further research is warranted to explore the factors influencing these differences and refine our understanding of diabetes manifestation in middle-aged adults.

Additionally, we explored the relevance and relative influence of symptoms and demographic factors in predicting diabetes in middle-aged adults. Our study's findings revealed a hierarchy of influential symptoms, with polyuria, polydipsia, alopecia, and irritability being the most critical indicators of diabetes among middle-aged individuals. Gender also played a significant role in this age group. This hierarchy provides valuable information for healthcare professionals, highlighting the crucial relevance of these symptoms in diagnosing diabetes among middle-aged adults. Notably, polyuria and polydipsia emerged as robust early signals for diabetes in this specific population. Conversely, symptoms such as itching, obesity, and muscle stiffness, although identified as important determinants in predicting diabetes among middle-aged adults, had a relatively low influence in predicting diabetes within this age demographic. Healthcare professionals can use this information to optimize diagnostic assessments, potentially expediting the identification of diabetes in middle-aged adults.

Precise classification of diabetes status holds paramount importance, as it prevents the progression of the condition and facilitates potential reversals, especially in cases of prediabetes. The recent surge in AI has spurred numerous research studies aiming to streamline the identification and diagnosis of diabetes, resulting in the need for models with exceptionally high accuracy measures. Several studies employing ML techniques for diabetes predictions consistently yield classification models with outstanding performance scores.^[40-46]

In our study, all the ML models exhibited commendable sensitivity scores, signifying their ability to accurately predict diabetes cases among the middle-aged adult population. Notably, the RF model stood out with the highest sensitivity score of 98.59%. The specificity scores were generally high and closely aligned, with the KNN algorithm emerging as the top performer in predicting non-diabetes cases among middle-aged adults, boasting a specificity score of 97.83%. Notably, the RF model excelled in overall accuracy and AUC, achieving scores of 96.58% and 96.00%, respectively. This aligns with the consensus in previous studies, highlighting the effectiveness of ensemble ML techniques in predicting or diagnosing diabetes across diverse populations.^[42,44,47-49]

This study has limitations, including its reliance on the dataset from India, which may affect the generalizability of results due to regional differences in healthcare and lifestyle. Additionally, given the secondary nature of our study's data, there was no information on the type of diabetes the patient had. As such, our study assumes that

all diabetes types share the examined symptoms, which may not be universally applicable. This potentially limits the clinical implications of the prevalent diabetes type in the dataset.

CONCLUSION

In conclusion, our study reveals key symptoms associated with diabetes in middle-aged adults, such as polyuria, polydipsia, gender, alopecia, irritability, and sudden weight loss. Additionally, it highlights the effectiveness of ML models, like the RF classifier, in predicting diabetes among middle-aged adults. Future studies should investigate genetic predispositions, lifestyle factors, and environmental influences to refine predictive models and develop personalized healthcare strategies, aiming to improve outcomes for at-risk individuals in this age group.

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Conflicts of interest

There are no conflicts of interest.

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